

GUARANTEES OF ORIGIN (GO) MARKET ANALYSIS AND PLAN TO SECURE OUR GREEN POWER SALES

April 2026



RESTRICTED



INTERNAL



SECRET



This study aims to anticipate tightening in Europe's green GO market

Conducted by Green Energy Acceleration Team

Objectives

In the coming years, the **European Green Guarantees of Origin (GO) market** (AIB) and **ENGIE own supply/demand balance** are facing **structural tightening**, driven by **rising demand, constrained supply, and possibly evolving regulations**.

In this context, the Green Energy Acceleration team conducted this study, that aims at analyzing potential GO deficit and proposing solutions to **secure our supply contracts**.

It focuses on six key European markets: France, Germany, Italy, Belgium, The Netherlands and Spain.

This study is part of the implementation of the audit recommendations regarding GO management and OneCert program.

Methodology

1. Estimate, for the coming years (2026 – 2030), the GO balance at both European and Country level → Modeled by [Market Analysis](#) & [Environmental Economics teams](#)
2. Calculate ENGIE's balance over the same timeframe in these markets → Calculated by [Green Energy Acceleration](#), [R&F](#), [BPs](#), [SPM](#) and [Carbon teams](#)
3. Propose an adapted pre-hedging strategy → Proposed by [EMA team](#)

European GO balance



Modeling **based on historical public data** on GO issuance and cancellation, with projections based on **Group scenarios**, considering a base case (notably regarding demand evolution and weather conditions).

Four scenarios were simulated:



1. Business as usual in **current regulatory conditions**
2. Implementation of mandatory **yearly matching** from 2028
3. **Stop in the use of Norwegian GO**, considered lower quality, to avoid reputational risk, respecting yearly matching from 2028
4. Use **of local GO only**, respecting yearly matching from 2028

ENGIE GO Balance



Calculation of ENGIE's balance:

ENGIE accessible GO

R&F available assets GO issuance data
(excl. CNR and Ocean Winds)



Contracted 3rd party Upstream PPAs
(registered in GEMforce)



Unbundled GO already sourced
(Provided by portfolio managers)

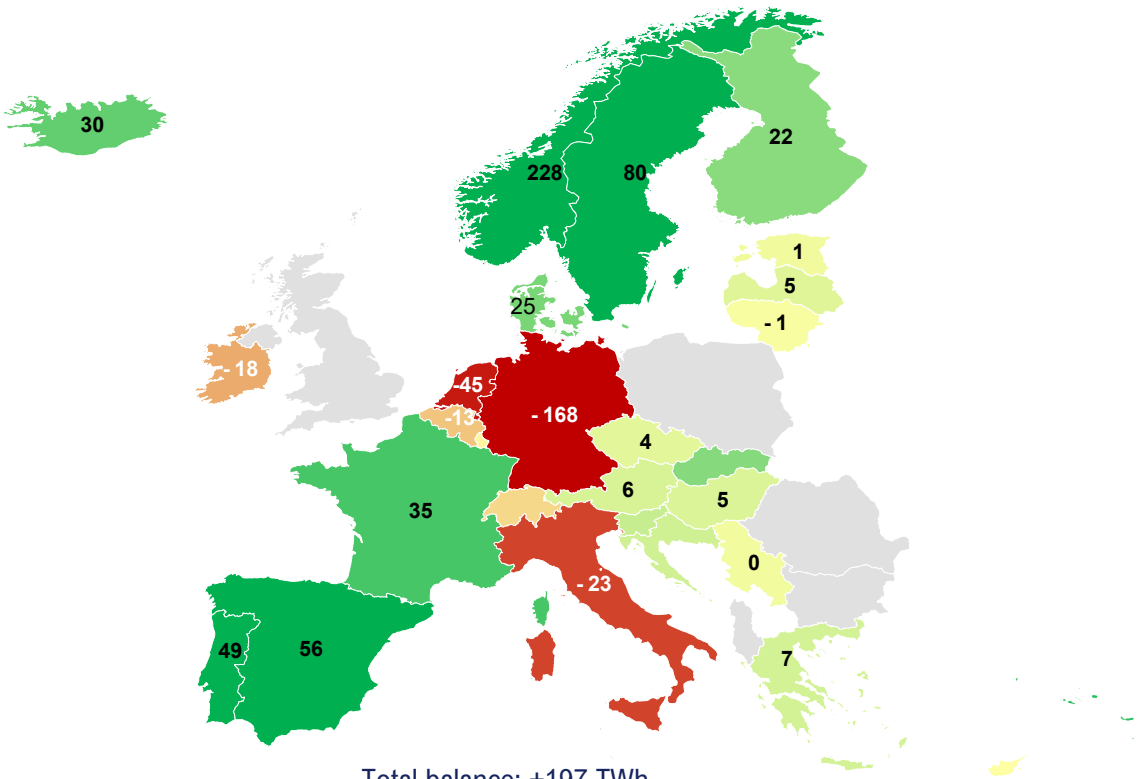
ENGIE GO demand

B2C, B2B and EM **MTP 2025 forecasts**

Within the current regulatory framework, the European GO market turns short by 2030, with wide disparities

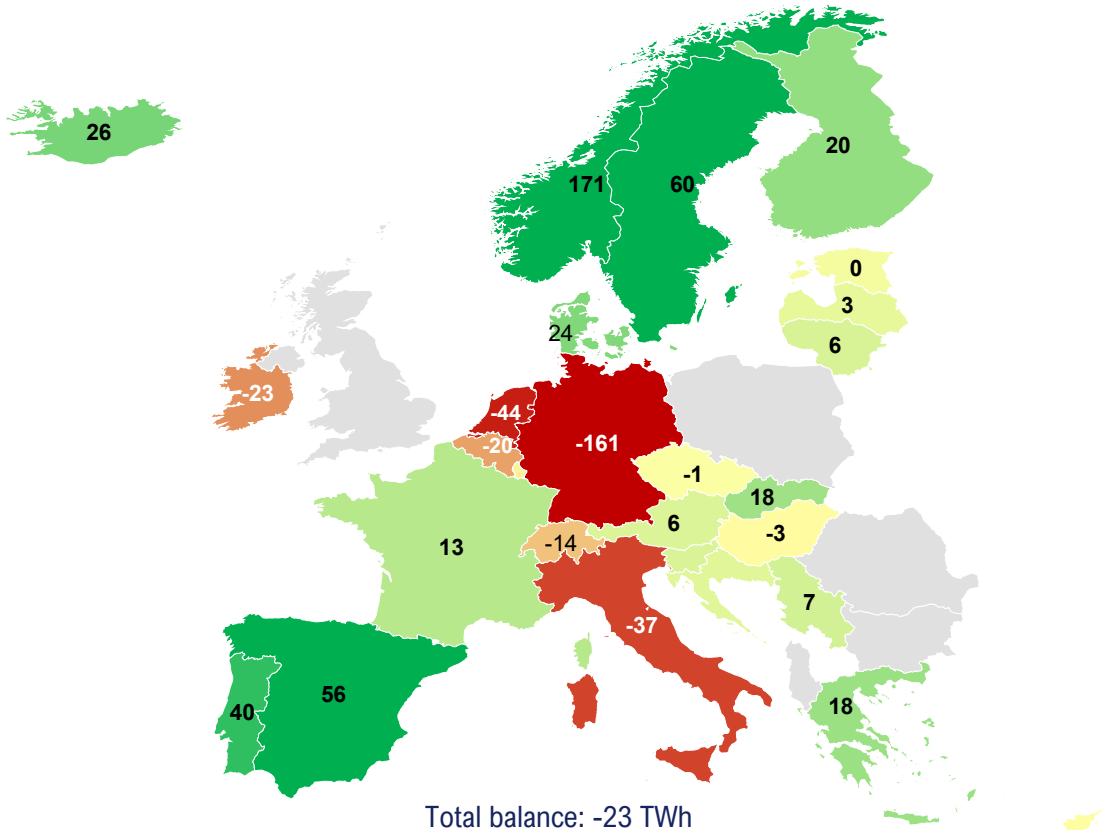
Source	Market Analysis model based on reviewed corporate scenario & AIB historical data
Assumptions	- Current regulations - Full country demand - Issuance per country - No cross-countries exchanges

2025 country balance
(Scenario 1, in TWh)



Total balance: +197 TWh
Total demand: 1 110 TWh

2030 country balance
(Scenario 1, in TWh)



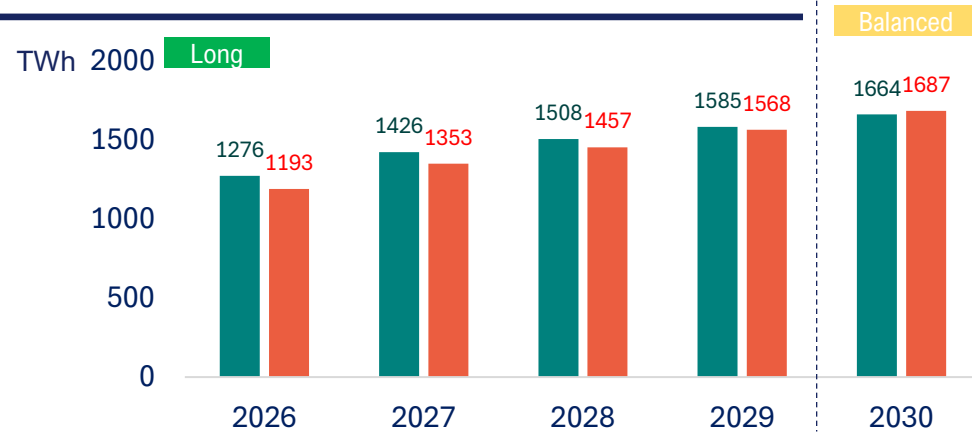
Total balance: -23 TWh
Total demand: 1 687 TWh

Legend: Deficit 0 Surplus Out of scope

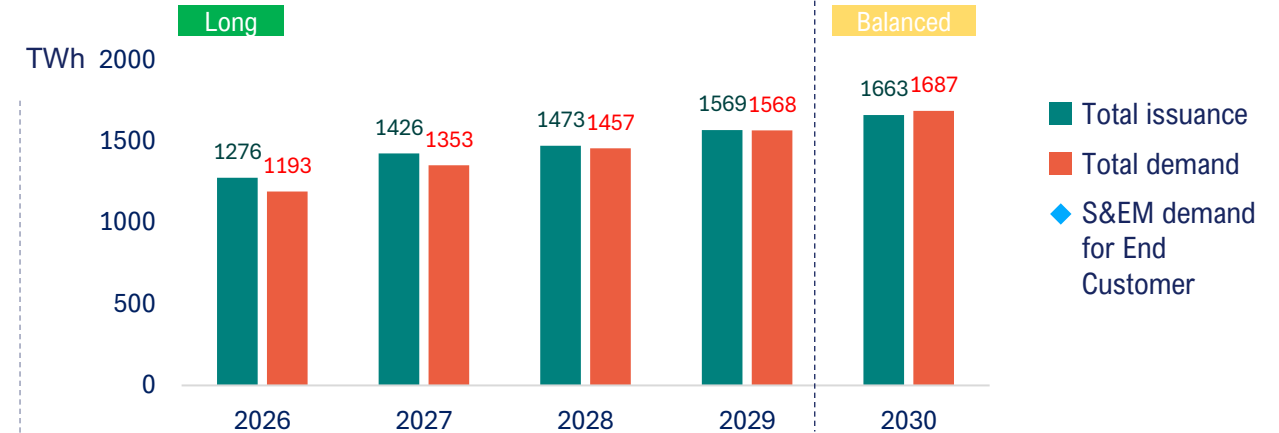
Note: Excludes atypical market participant behaviour

Country Balance = Issuance – Demand + Rollover Y-1
Rollover Y-1 = Issuance Y-1 – Demand Y-1

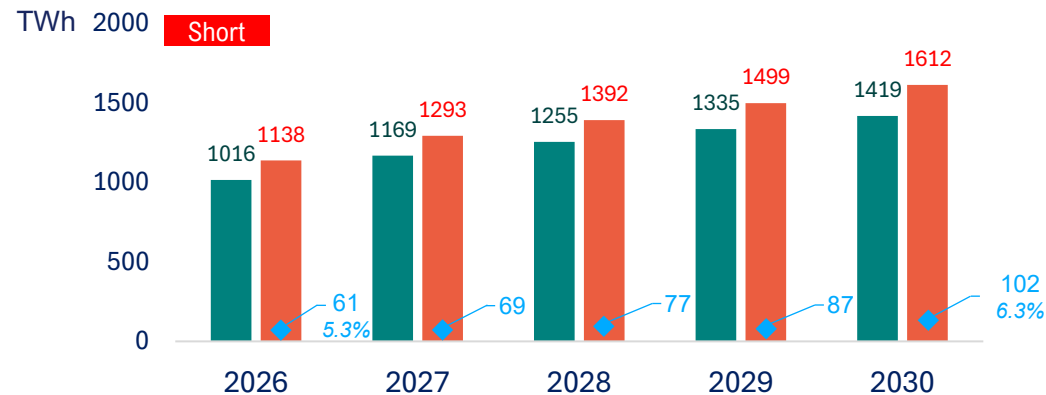
European market length relies on low-quality GO, while ENGIE-country market remains short



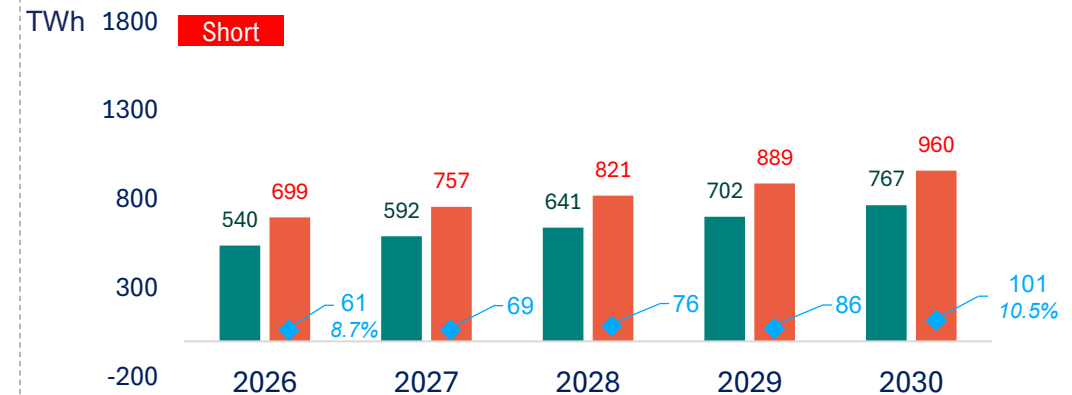
Scenario 1: Current situation.
Scope: All AIB Countries – No change in regulation – No yearly matching (excl. For FR and ES)



Scenario 2: Regulation imposes yearly matching from 2028.
Scope: All AIB Countries – Yearly matching from 2028



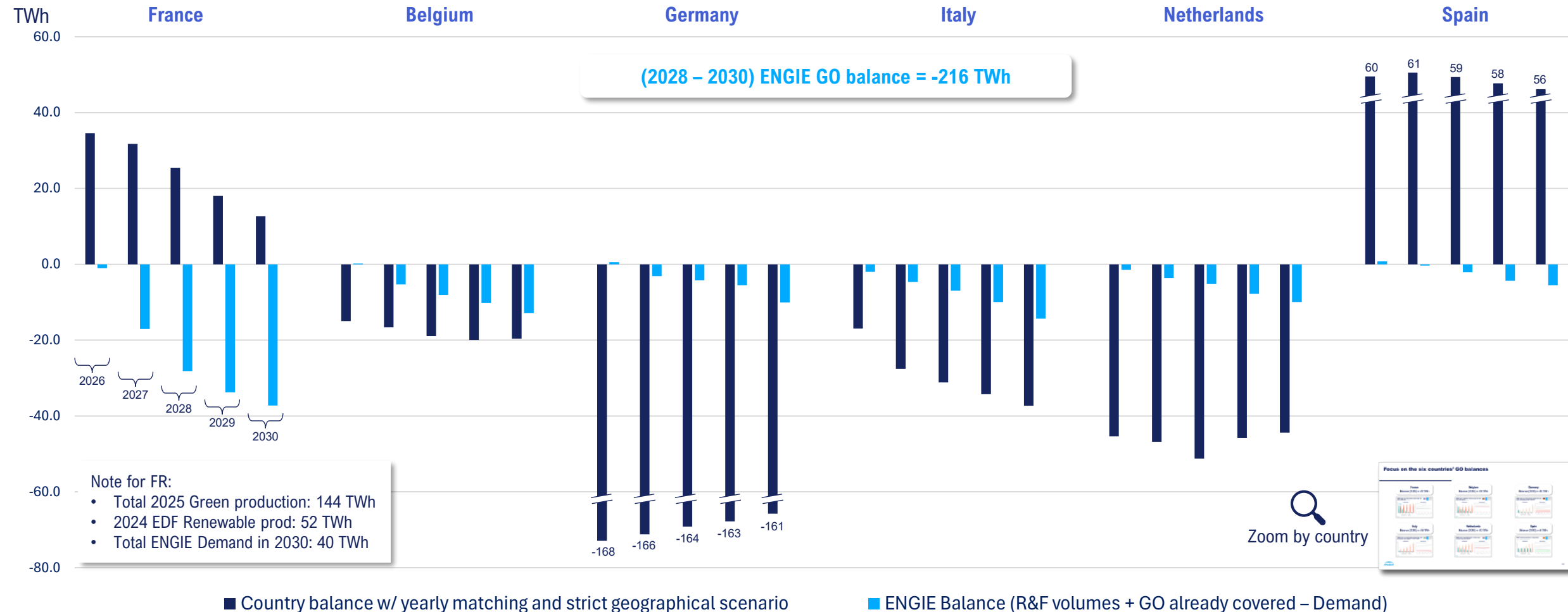
Scenario 3: ENGIE and main market players are stopping their sourcing of Norwegian GO to avoid greenwashing accusation.
Scope: AIB countries, excl. Norway – Yearly matching from 2028



Scenario 4: Regulation imposes strict geographical and yearly matching from 2028.
Scope: ENGIE countries only (FR, BE, DE, IT, NL, ES)

Market length vs. ENGIE shortage in our countries

Scenario 4 - Yearly matching - Local GO only

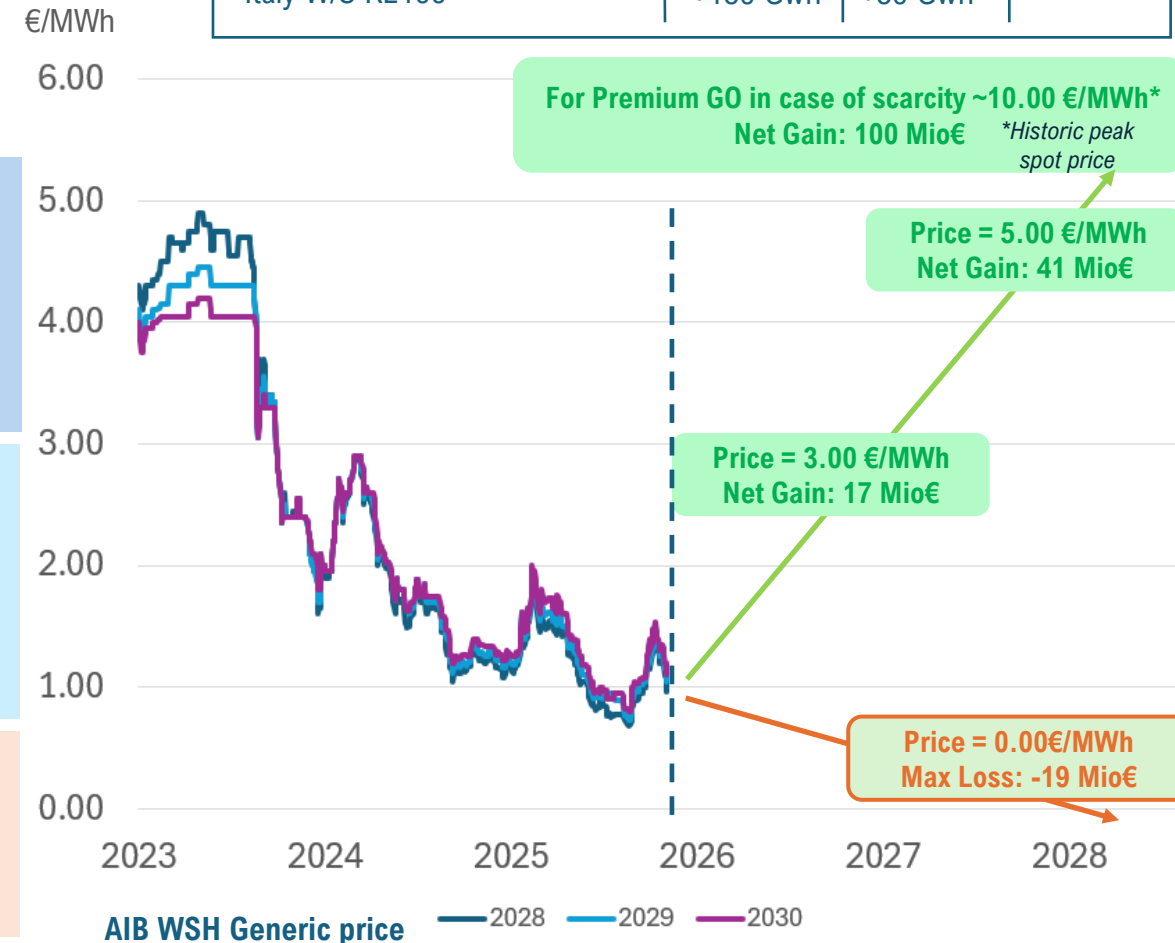


We propose a hedging acceleration in 2026 to start securing 2028-2030 GO, relevant to our green power ambitions

We propose to secure **4TWh/year** - up to **12 TWh (~5% of our deficit)** for a total **max budget of ~19Mio€**

Already secured by EMA	2028	2029	2030
France GO Monthly Profile	+850 Gwh	+850 Gwh	
France GO 24/7			+300 Gwh
Italy W/S RE100	+150 Gwh	+50 Gwh	

	Targeted volume: of the ENGIE gap	Average Price 2028-2029-2030	Max Budget
Goal 1 - Targeted Countries: France. Germany. Italy. Spain. BE. NL - Asset Specific - Technology Wind / Solar - Granularity: Monthly profile with hourly load curve	Min 1.3 TWh/Yr = 4TWh Total	1.78 €/MWh (+0.60€/MWh premium vs AIB)	7.1 Mio€
+			
Goal 2 - Targeted Countries: France. Germany. Italy. Spain. BE. NL - Technology Wind / Solar - Production period: Monthly profile (French supply)	Min 1.3 TWh/Yr = 4TWh Total	1.57 €/MWh (+0.40€/MWh premium vs AIB)	6.3 Mio€
+			
Goal 3 - Targeted Countries: France. Germany. Italy. Spain. BE. NL - Technology Wind / Solar - Asset COD < 15Y	Min 1.3 TWh /Yr = 4TWh Total	1.37 €/MWh (+0.20€/MWh premium vs AIB)	5.5 Mio€





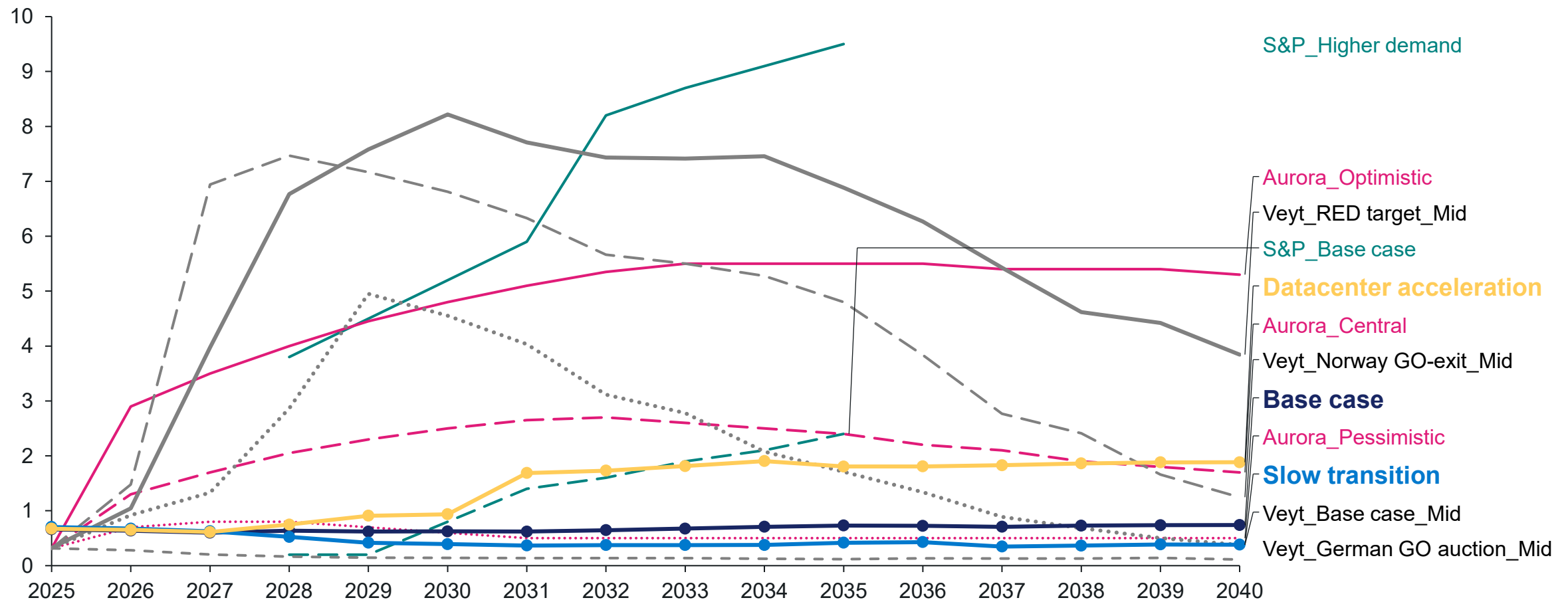


Appendices

Prices – benchmark base case scenario with external sources

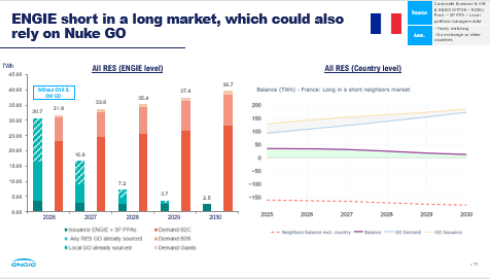
Wide range of prices illustrate complexity of EU GO markets – no consensus emerges

Market Analysis base case price vs. S&P, Aurora & Veyt scenarios – €₂₀₂₄ / MWh

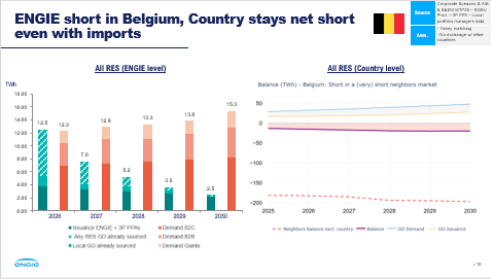


Focus on the six countries' GO balances

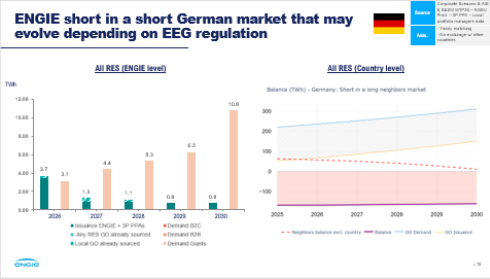
France
Balance (2030) = -37 TWh



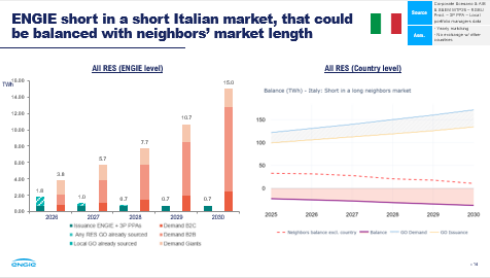
Belgium
Balance (2030) = -13 TWh



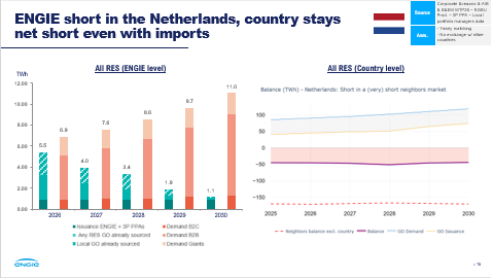
Germany
Balance (2030) = -10 TWh



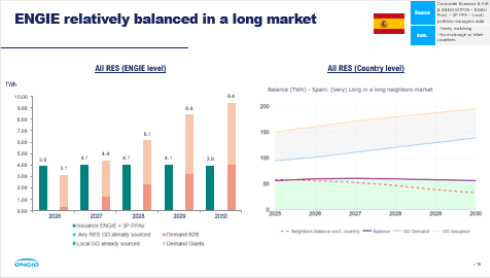
Italy
Balance (2030) = -14 TWh



Netherlands
Balance (2030) = -10 TWh



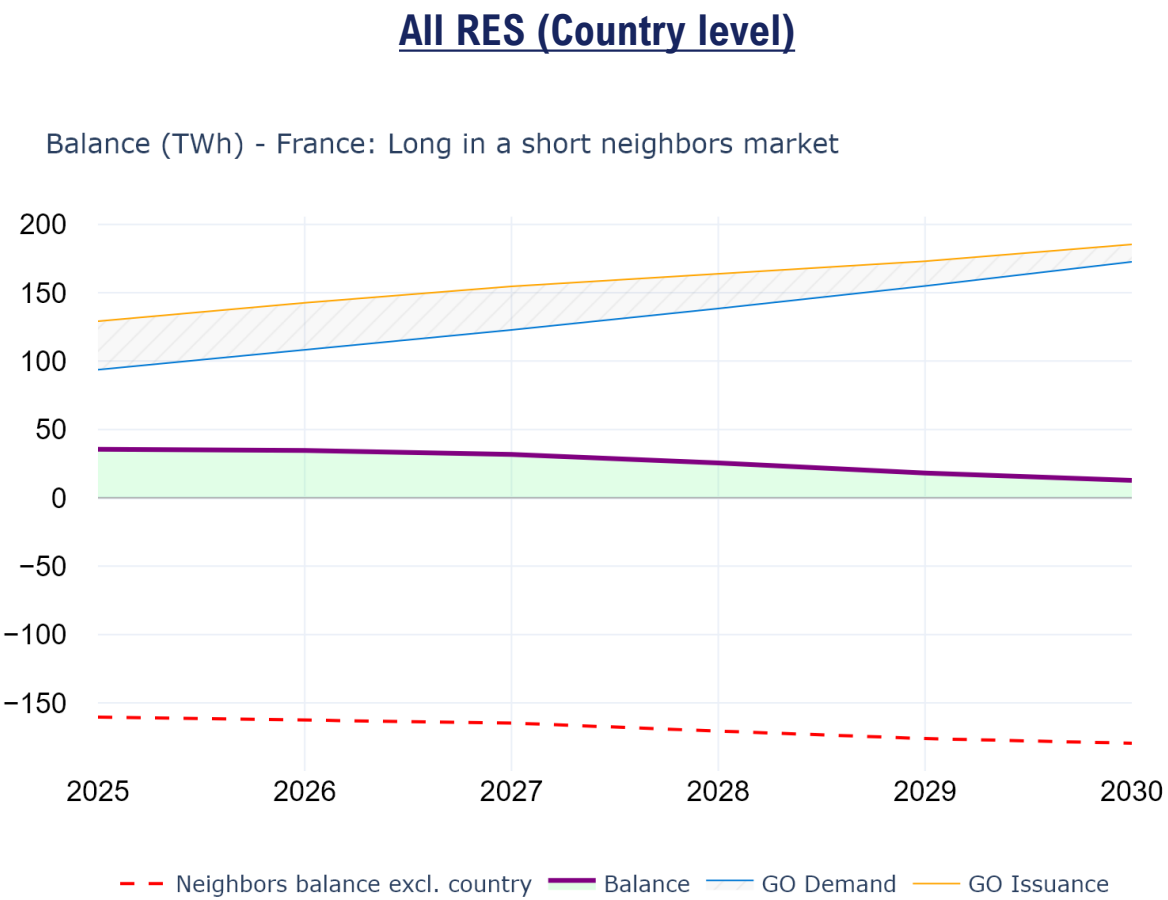
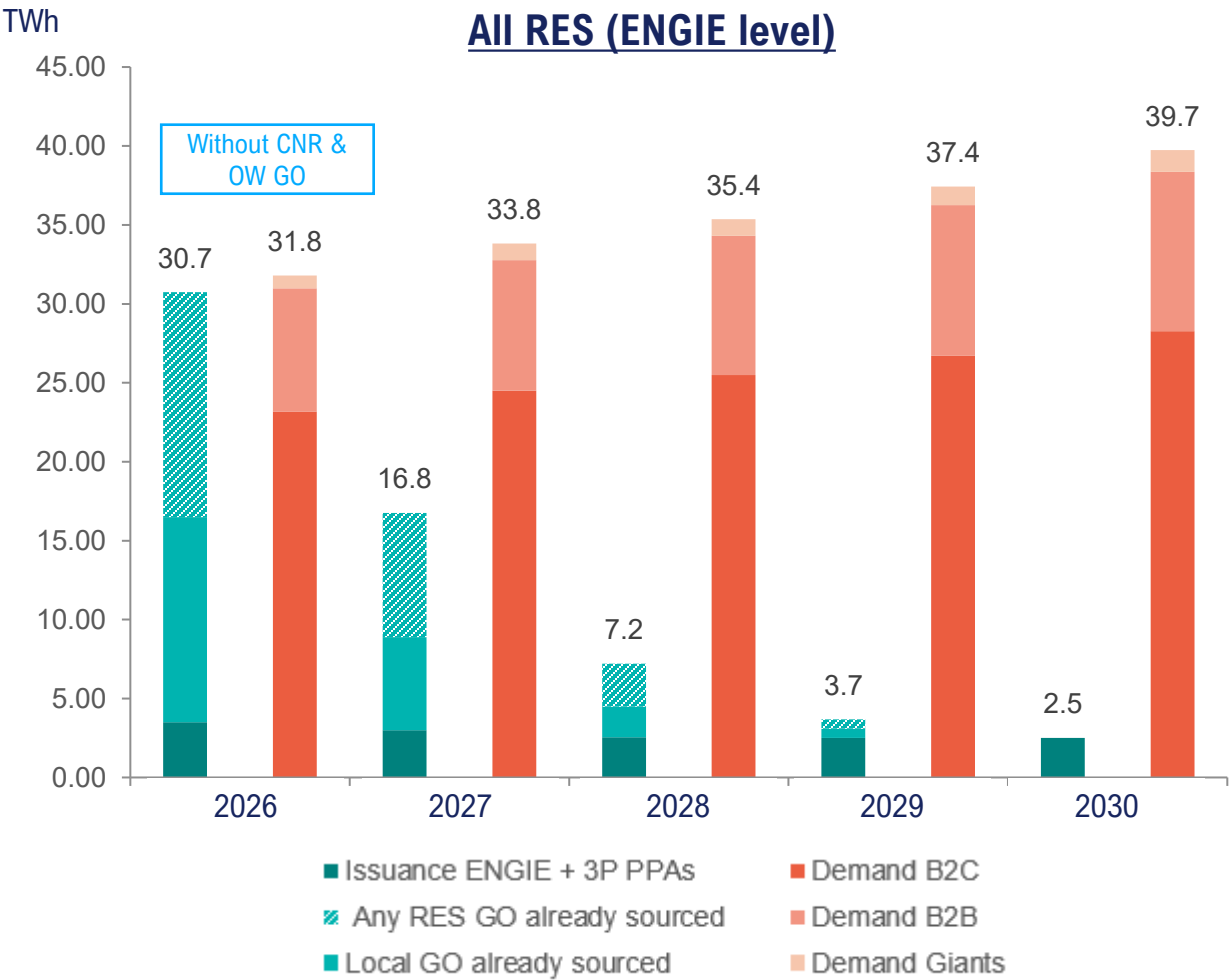
Spain
Balance (2030) = -6 TWh



ENGIE short in a long market, which could also rely on Nuke GO



Source	Corporate Scenario & AIB & S&EM MTP25 – RGBU Prod. – 3P PPA – Local portfolio managers data
Asm.	- Yearly matching - No exchange w/ other countries

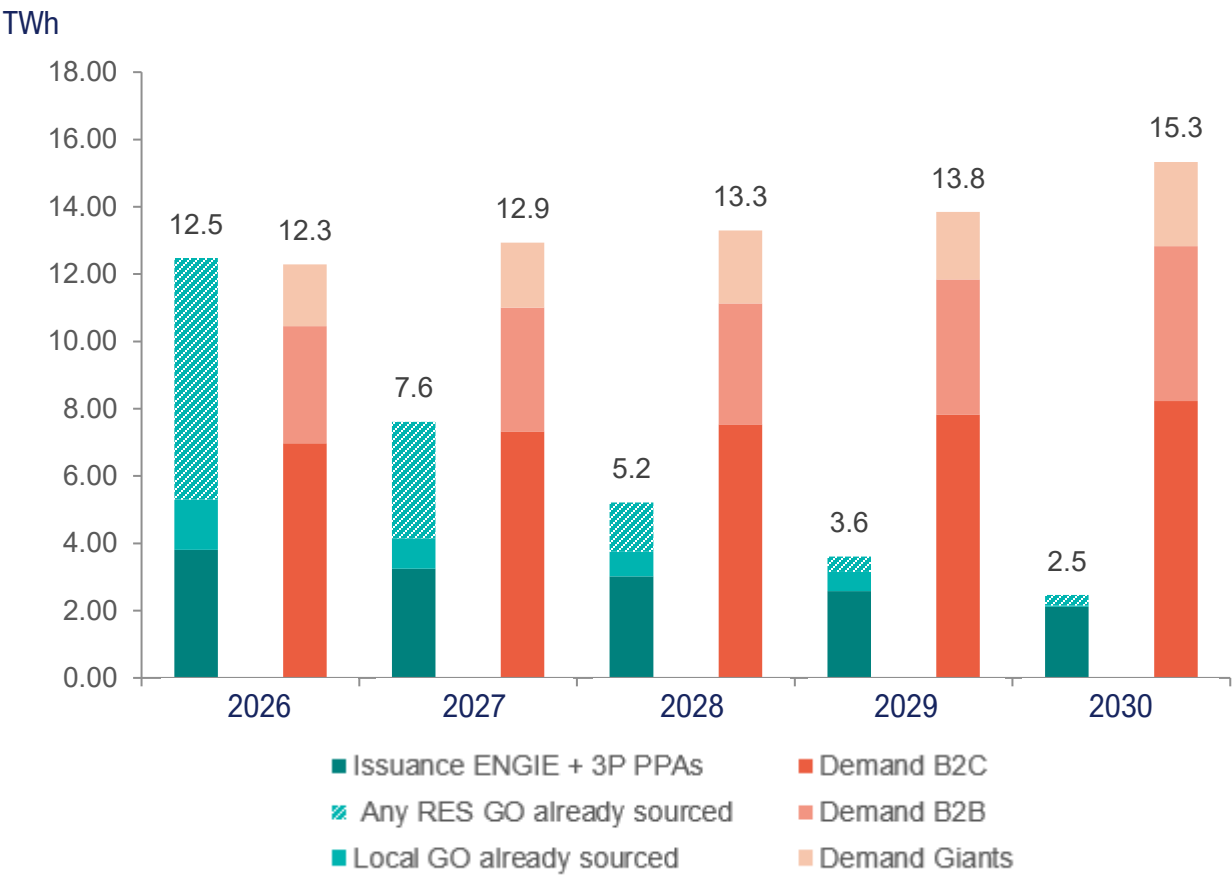


ENGIE short in Belgium, Country stays net short even with imports

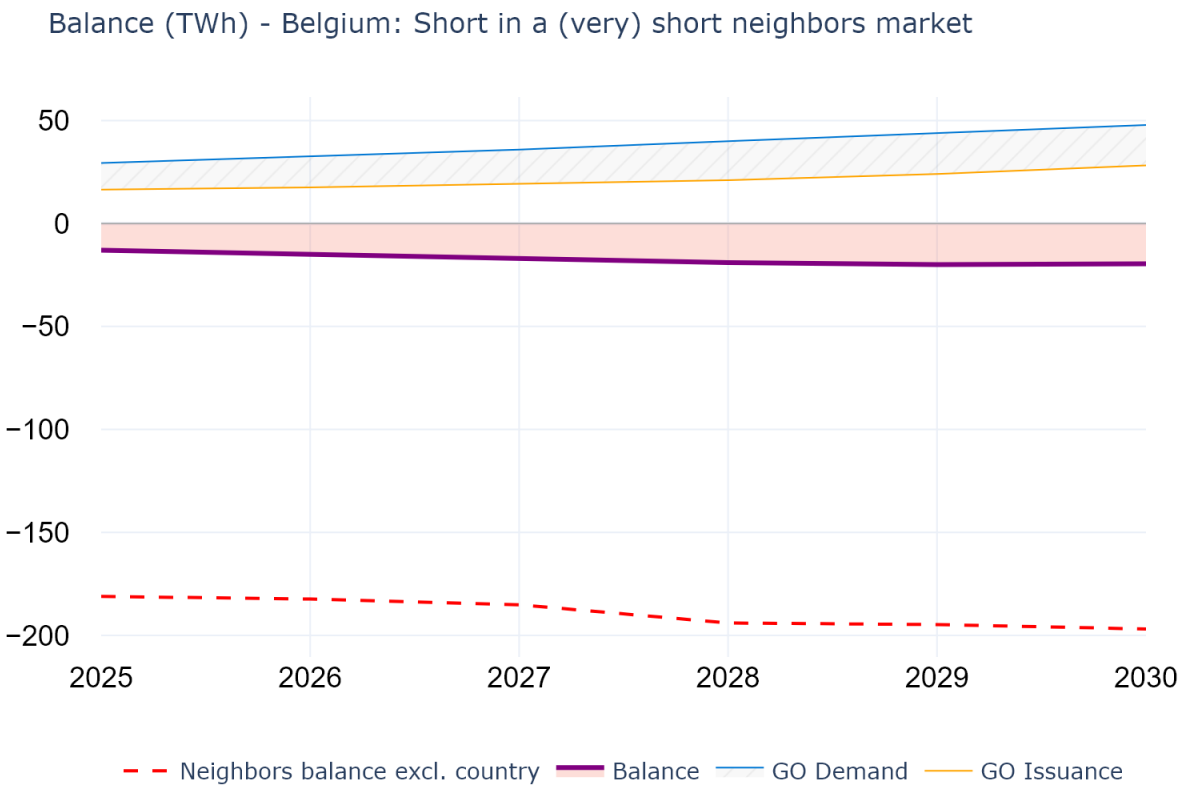


Source	Corporate Scenario & AIB & S&EM MTP25 – RGBU Prod. – 3P PPA – Local portfolio managers data
Asm.	- Yearly matching - No exchange w/ other countries

All RES (ENGIE level)



All RES (Country level)



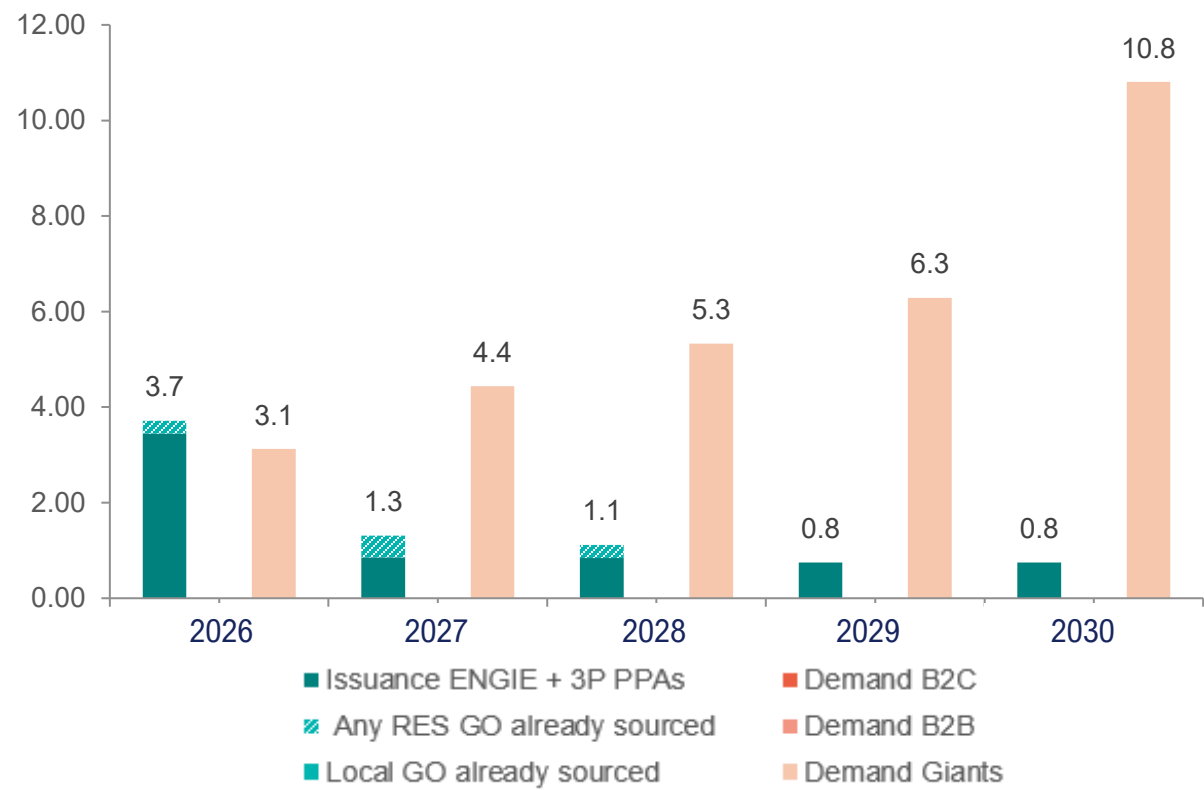
ENGIE short in a short German market that may evolve depending on EEG regulation



Source	Corporate Scenario & AIB & S&EM MTP25 – RGBU Prod. – 3P PPA – Local portfolio managers data
Asm.	- Yearly matching - No exchange w/ other countries

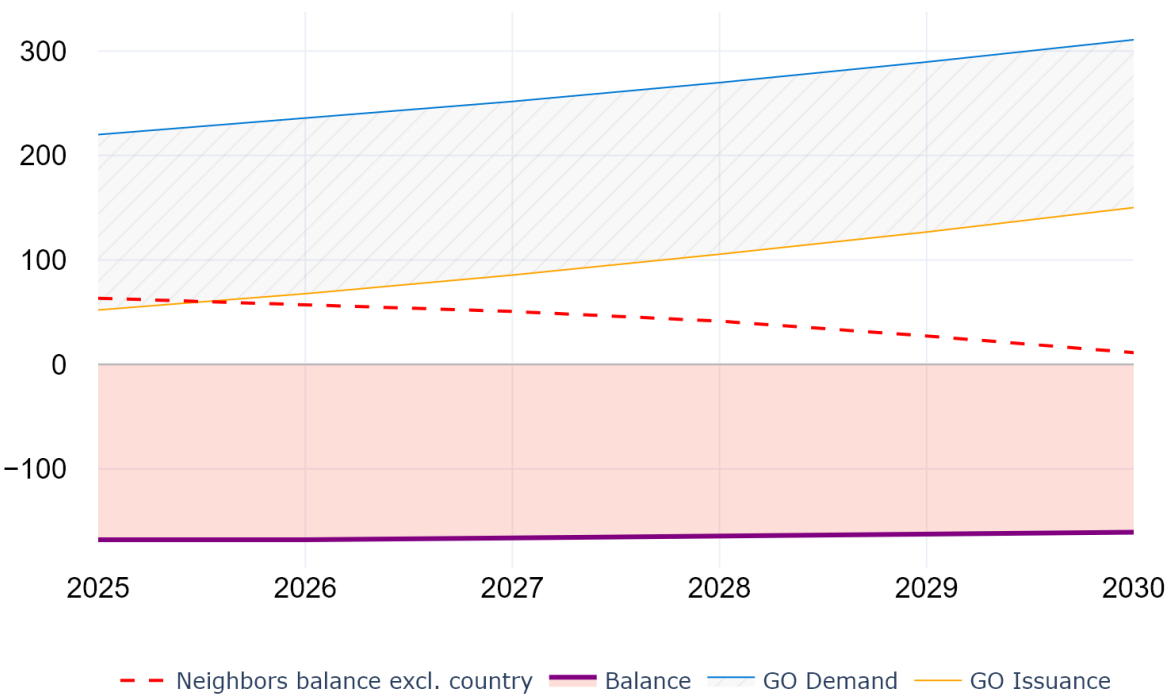
All RES (ENGIE level)

TWh



All RES (Country level)

Balance (TWh) - Germany: Short in a long neighbors market

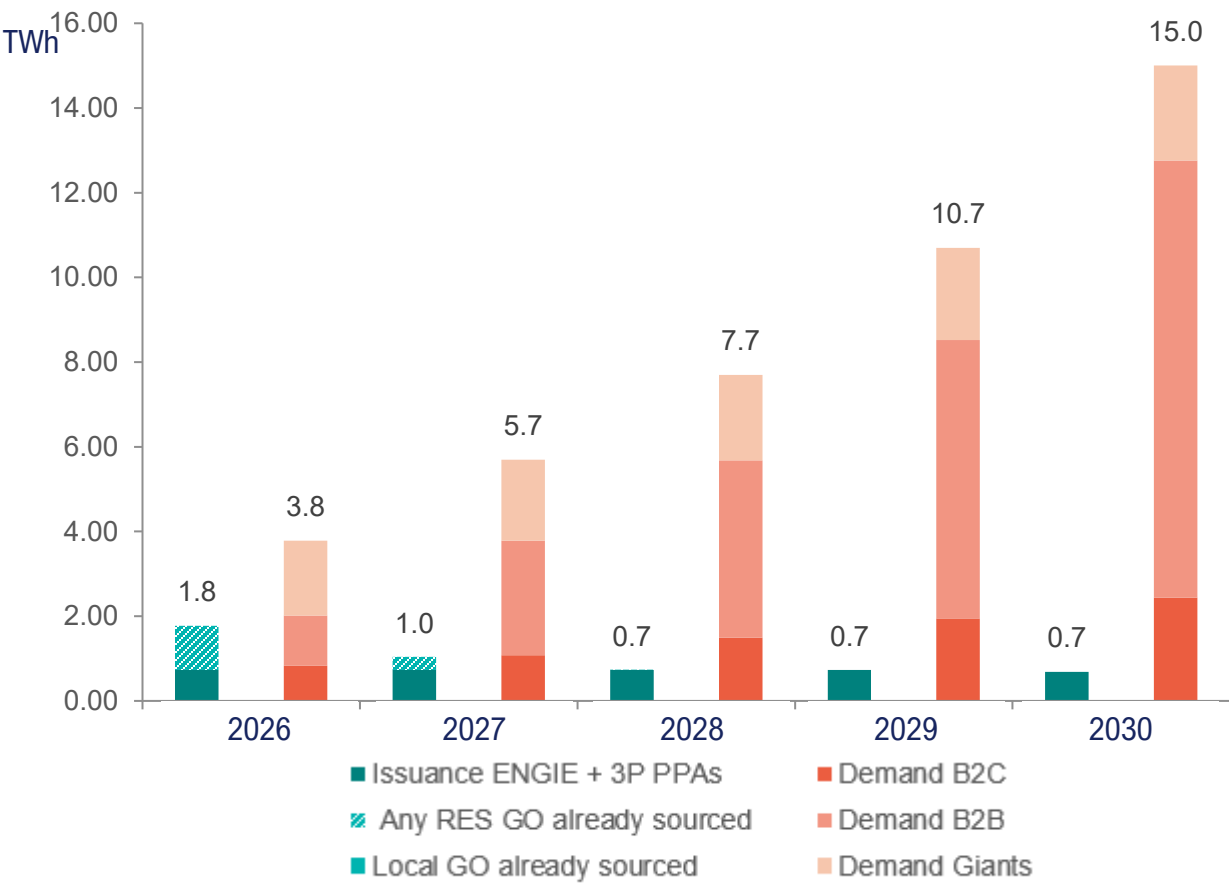


ENGIE short in a short Italian market, that could be balanced with neighbors' market length

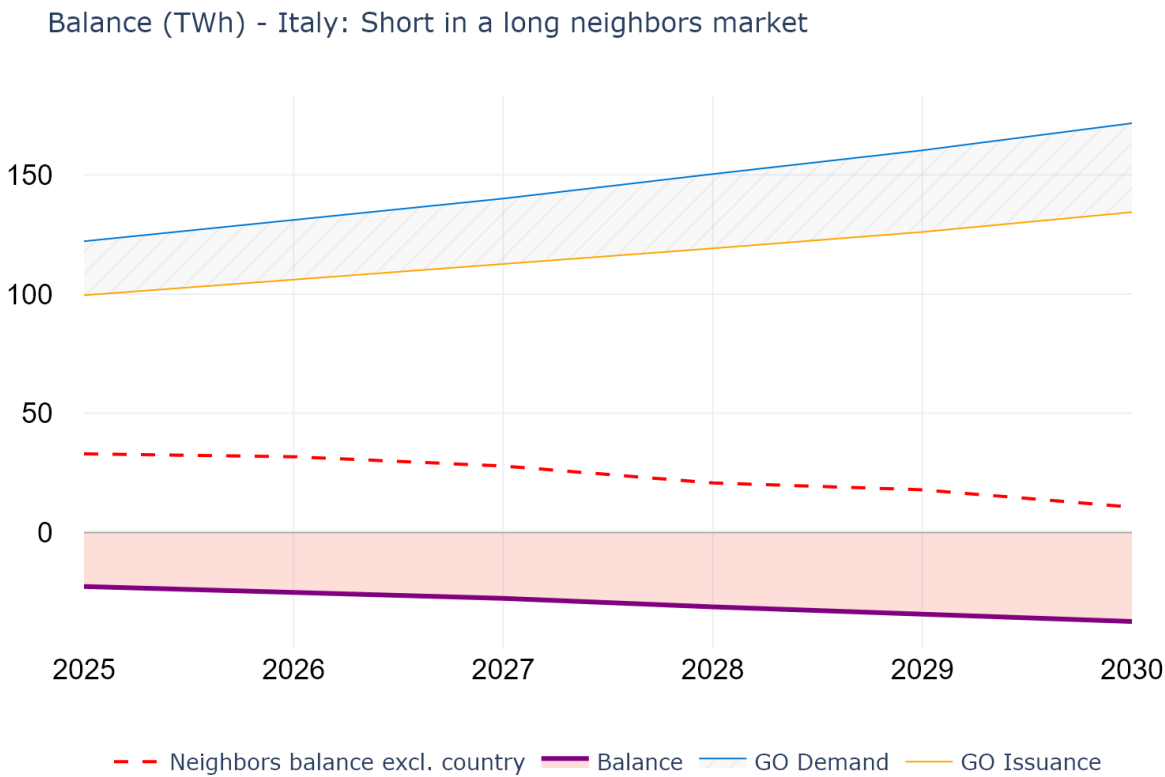


Source	Corporate Scenario & AIB & S&EM MTP25 – RGBU Prod. – 3P PPA – Local portfolio managers data
Asm.	- Yearly matching - No exchange w/ other countries

All RES (ENGIE level)

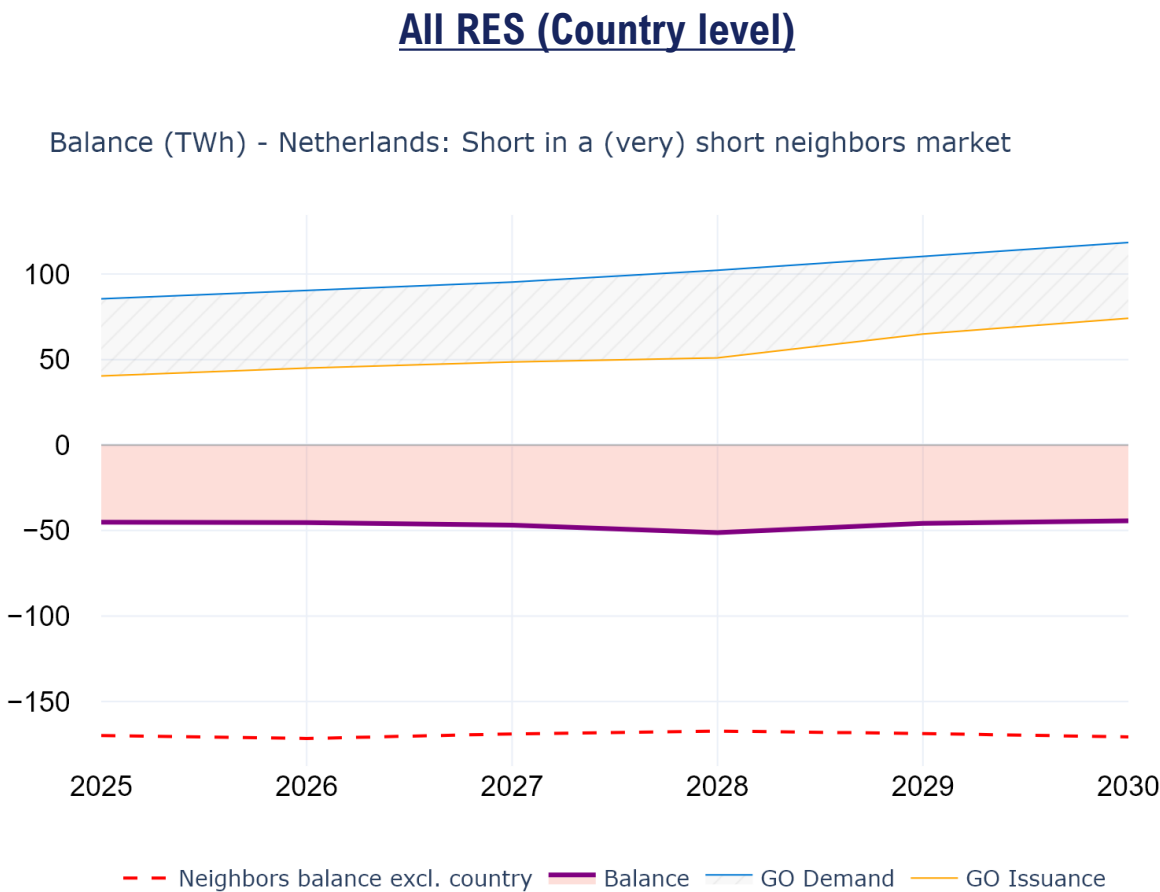
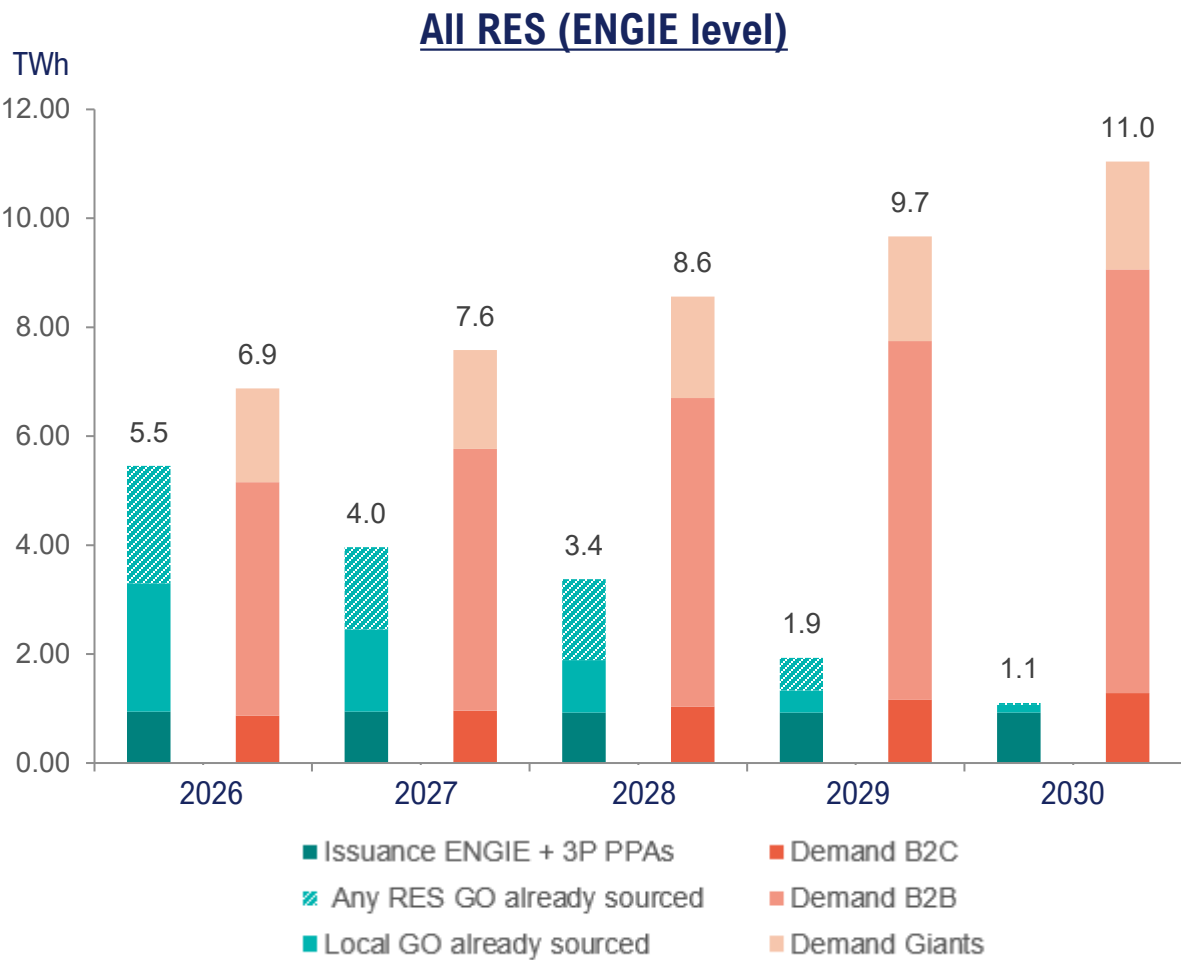


All RES (Country level)



ENGIE short in the Netherlands, country stays net short even with imports

	Source	Corporate Scenario & AIB & S&EM MTP25 – RGBU Prod. – 3P PPA – Local portfolio managers data
	Asm.	- Yearly matching - No exchange w/ other countries

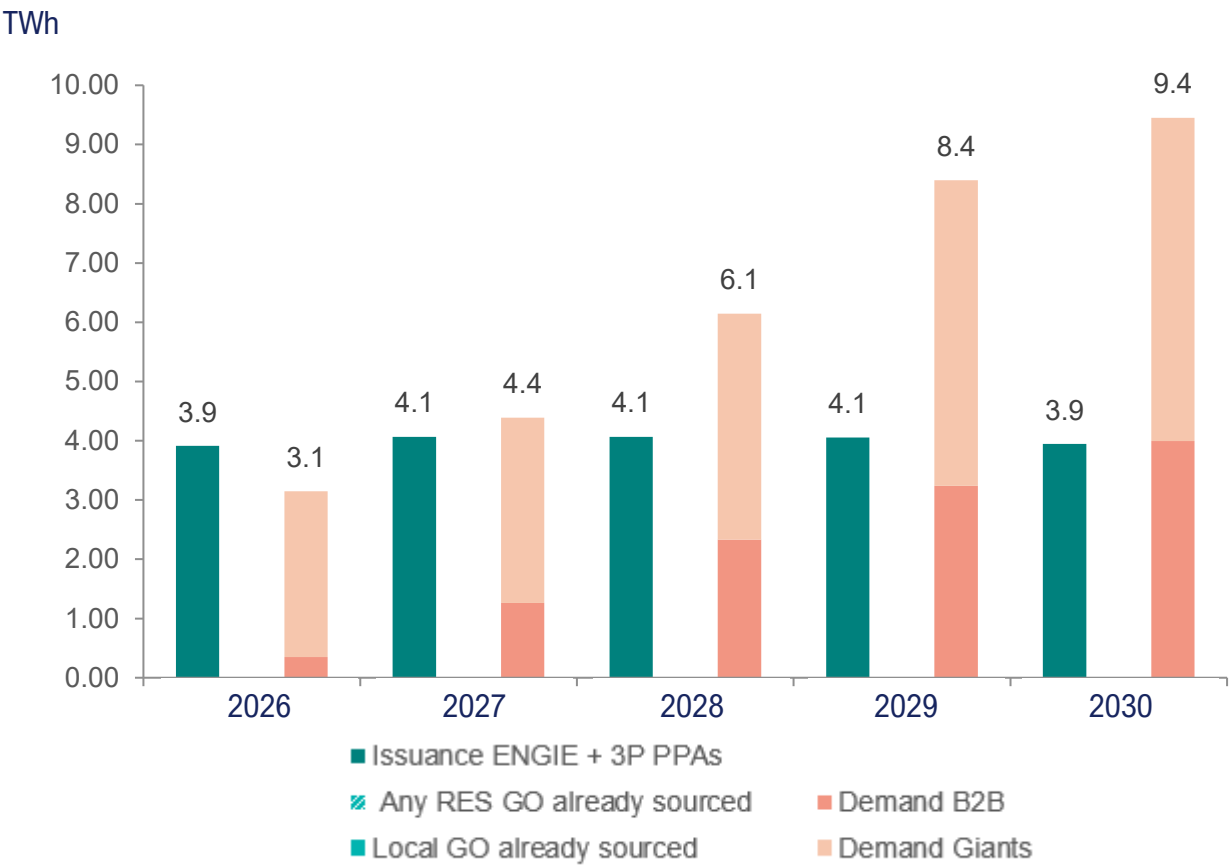


ENGIE relatively balanced in a long market

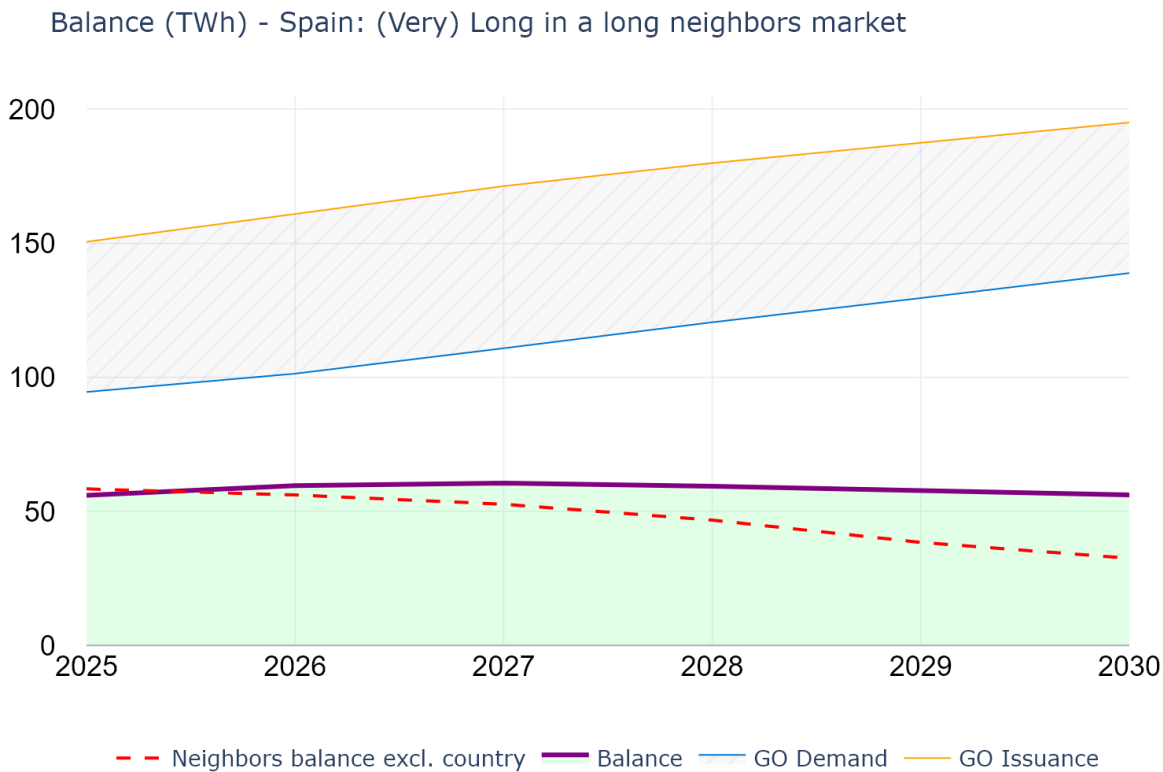


Source	Corporate Scenario & AIB & S&EM MTP25 – RGBU Prod. – 3P PPA – Local portfolio managers data
Asm.	- Yearly matching - No exchange w/ other countries

All RES (ENGIE level)



All RES (Country level)



ENGIE GO balance per country

Source

S&EM MTP25
R&BU Prod. – 3P PPA –
Local portfolio managers
data

Country					
	2026	2027	2028	2029	2030
France	-1.1	-17.1	-28.1	-33.7	-37.2
Belgium	0.2	-5.3	-8.1	-10.2	-12.9
Germany	0.6	-3.1	-4.2	-5.5	-10.1
Italy	-2.0	-4.7	-7.0	-10.0	-14.3
Netherlands	-1.4	-3.6	-5.2	-7.7	-9.9
Spain	0.8	-0.3	-2.1	-4.3	-5.5
Total	-3.9	-34.3	-54.7	-71.6	-89.9

